MUSIC EMOTION RECOGNITION BASED ON LYRIC AND AUDIO FEATURES

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Writing Sample

ABSTRACT

In this paper, we conduct music emotion recognition (MER) based on the lyrics and the audio signals of a music clip. We adopt the continuous Valence-Arousal as the descriptor of music emotion, and we propose a multi-modal deep learning model which can process the static features and dynamic audio features at the same time by dividing the model into the static part and dynamic part, and we adopt attention mechanism to combine the predictions of these two parts by gating networks. Our model is versatile and flexible as it is capable of recognizing the global emotion as well as the dynamic emotion of music per second, which is another innovation. Besides, we use new methods to extract features for MER, such as using pretrained models like BERT and musicnn. We conduct experiments on the public PMEmo dataset, and show that our model has achieved significant improvements.

$$Y_i \equiv Y_i(W_i) = \begin{cases} Y_i(1) & \text{if } W_i = 1 \text{ (treated)} \\ Y_i(0) & \text{if } W_i = 0 \text{ (control)} \end{cases} \tag{1}$$

$$\tau := \mathrm{E}[Y_i(1) - Y_i(0)]. \tag{3.1}$$

$$Y_i(1), Y_i(0) \perp W_i. \tag{3.2}$$

$$\tau(x) := \mathrm{E}[Y_i(1) - Y_i(0)|X_i = x], \tag{4.1}$$

$$\mathrm{E}[Y_i(1) - Y_i(0)|G_i = g], \tag{4.2}$$

1. INTRODUCTION

Music emotion recognition (MER) has become a research hotspot in recent years. As emotion provides semantic information of music, it is the essence feature of music [1]. Music emotion recognition has lots of applications, in recent years, personalized music recommendation systems based on machine learning are very popular, and in order to achieve emotion-based music retrieval, it is necessary to label the emotions of music works. Therefore, research on MER technology to analyze the audio signal of music [2] and to realize automatic emotion recognition of music, has become urgent.

The research of MER can be traced back to 1988 [3]. Since multimedia data cannot be directly matched with human emotion perception, it is necessary to analyze the association between human emotion and the feature of data. Music emotion classification and recognition are mainly based on the two differences features forms: audio signal and lyrics text. For audio signals, researchers have found that features such as music beats and rhythms based on machine learning methods can be used to identify music emotions; for lyrics text, researchers have made some progress using the Bag-of-Words method, and emotion words are also commonly used to retrieve and describe music [4].

In this paper, we focus on predicting the global emotion and dynamic emotion of music as measured by valence and arousal using features extracted from the audio and lyric of the music. Current researches mainly recognize music's global emotion, that is, the overall emotion of the music, however, emotion can vary a lot in different episodes of a single music. Therefore, it makes more sense to predict the dynamic emotions of music, which is more difficult and requires more refined methods. There are few researches on the dynamic recognition of music emotion, and fewer on recognition of both global and dynamic emotion. Among them, most of the researchers build two separate models for global and dynamic recognition tasks, which lack flexibility. We propose a single model which can recognize the global emotion as well as the dynamic emotion of each second of the music, by adding an extra learnable embedding to the original music sequence embedding, and utilize the sequential nature of Transformer to predict the global and dynamic emotion of music clips.

Traditional MER models utilize features extracted from the original audio, such as Log-Mel Spectrogram, Mel-Frequency Cepstral Coefficients, and linear predictive cepstral coefficients. There are some researches on the multi-modal recognition of music emotion, such as combining audio features with text features from lyrics [5] [6] or the listeners' electrodermal activity (EDA) signals [7]. However, most of these multi-modal methods use a quite coarse way to utilize features from different modalities, such as simply concatenate features from different sources and input them to a single machine learning or deep learning model. These methods are insufficient to capture the complex associations between different modalities, and have poor performance in differentiating the importance of modalities on different emotion dimensions. Instead of directly concatenate features from different modalities, we

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introduce modal heads which are linear projections applied on features from difference sources, and we concatenate the features after the projection. Moreover, we divide our model into two parts, one is the static part which mainly inputs text features and the other is the dynamic part which inputs dynamic audio features, and uses Transformer to learn the inner sequential relationship of audio features. We introduce gating network based on attention mechanism to aggregate the predictions from the static part and the dynamic part. The main contributions of our work are as follows:

- We propose a deep learning model which can recognize the global emotion and dynamic emotion of music at the same time.
- We better utilize features from different sources by introducing modal heads and use gating networks to aggregate predictions of the static part and the dynamic parts of the model.
- We compare our model with other models on the public PMEmo dataset [8], and demonstrate that our model has the best performance. We conduct extensive experiments, such as change Transformer in the dynamic part to RNN, LSTM or GRU, and we also analyze the importance of different sources of features and the importance of the static and dynamic parts for valence and arousal recognition.

The rest of the paper is organized as follows. We discuss related work of music emotion recognition in Section 2. We propose the lyric and audio features that we consider useful for music emotion recognition in Section 3. The detail of our model is presented in Section 4. The experiments are shown in Section 5. Finally, brief conclusions are made in Section 6.

2. RELATED WORK

2.1 Music Emotion Recognition

The purpose of music emotion recognition (MER) is to identify the emotion of unknown music through the existing music database, that is, to build a model with the help of existing experience. Therefore, the MER problem can be abstracted as a pattern recognition problem, that is, a machine learning problem, which can be solved using machine learning algorithms.

In MER, emotions are represented as discrete single-label or multi-label models, and continuous emotional space models. By analyzing multi-modal music features such as content text and acoustic laws, we can predict music emotions.

2.1.1 Discrete Emotion Model

In 1936, Hevner [9] proposed a discrete category of music emotional model, using 67 emotional adjectives to describe the emotional space expressed by music, and divided these 67 emotional adjectives into 8 categories, each category has a representative adjective.

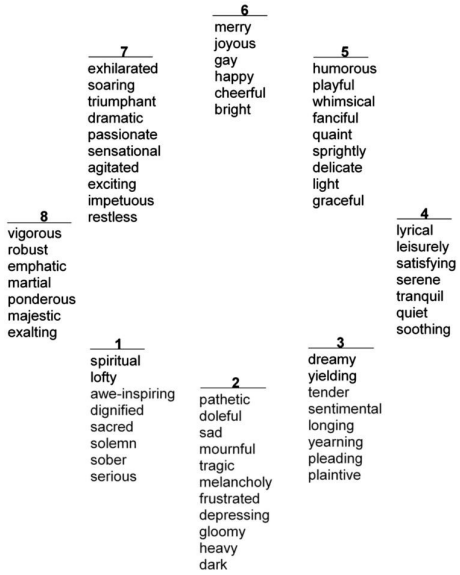

Figure 1. Overview of Hevner circle

The use of this model highlights the bipolarity of music emotions (such as happiness/sadness), and represents their possible ways across circles in space, and shows the multi-level and multi-label nature of music emotion classification. The Hevner model is widely used in related research in the field of music, and researchers continue to improve the Hevner emotion model, such as UHM9 (Update of Hevner's adjective Model (UHM) in nine categories), 5BE (5 basic emotions(Happy, Sad, Tender, Scary, Angry)), 4BE(4 base emotions(Happy, Sad, Angry, Fearful)).

2.1.2 Continuous emotion model

In addition to music discrete category emotion models, some scholars have proposed that music emotions are expressed through continuous spatial dimensions. The most famous is the Valence-Arousal emotion model proposed by Russell [10] (V-A model) . The model believes that the emotional state is a point distributed in a two-dimensional space containing Valence (valence) and Arousal (activation). The valence reflects the degree of positive and negative emotions, and the degree of activation reflects the intensity of emotions. Based on this representation, any emotional state can be represented as a point on the VA coordinate plane. That is, each point in the coordinate plane represents an emotional state. As shown in Figure 2. In this experiment, the VA model is selected to characterize the music dimension.

2.2 Machine Learning Models for MER

In the early days, SVM (Support Vector Machines) [11] and BPNN (Backpropagation Neural Network) [12] were often used for single-label emotion recognition, however, their accuracy were not very good. Subsequently, in the multi-label research of MER, some literature compared binary correlation (BR), label Powerset (LP), random k-label set (RAkEL) and multi-label k-nearest neighbor (MLkNN). For comparison, it is found that RAkEL

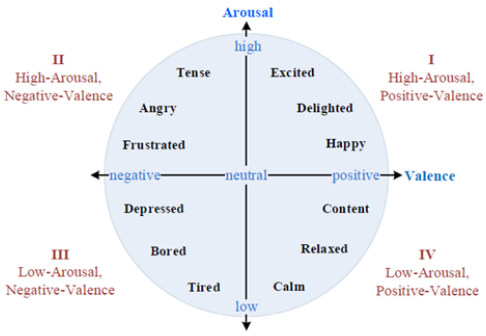

Figure 2. VA Model

had better performance and can reach an accuracy of 79%, which was better than the best recognition result in the AMC task of MIREX. In the research of MER's dimensional emotional space model, support vector regression (SVR), Adaboost, RT and Gaussian Mixture Model (GMM) classifiers had good recognition results.

2.3 Deep Learning Models for MER

Weninger [13] created a LSTM network with a single hidden layer trained to predict the input features, i.e., $y_t = x_t$. To avoid over-fitting, in each training epoch and timestep t, they added a noise vector n to x_t which is sampled from a Gaussian distribution with zero mean and variance σ_n. Sequentially, Chaki [14] described three attentive LSTM based approaches for dynamic emotion prediction from music clips. Gao [15] then used CRNN (CNN+RNN model) to classify emotions in simple melody music. The CRNN model takes the two-dimensional vector extracted from the audio according to a fixed structure in advance as input. Then, in the RNN substructure, they take advantage of the ability of RNN model to extract time-series features, the two-dimensional feature vector output from the CNN substructure is then input into RNN model, and finally, a one-hot vector is obtained through a fully connected layer.

Yin [7] proposed Res-SIN, an E2E framework using EDA signal images to classify human emotion, combining individual emotion features and external emotion benchmarks to accelerate convergence. Res-SIN applies convex optimization-based EDA (cvxEDA) to decompose signals and mine the static and dynamic emotion changes, then transforms decomposed signals to images for CNN frameworks. Furthermore, Yin [16] proposed an E2E multimodal framework, the 1-dimensional residual temporal and channel attention network (RTCAN-1D), focusing on the fusion of the subject individual EDA features and the external evoked music features. Subsequently, a convolutional neural network built around VGGNet and a novel post-processing technique were proposed by Sarkar [17]. Proposed methodology provides substantial improvement of performance for the datasets.

2.4 Pretrained Audio Models

Audio pattern recognition is crucial in the machine learning area, and it includes a myriad of tasks such as audio tagging, music classification, sentiment analysis, and audio generation. However, compared with image datasets, most audio datasets have lots of defects: limited duration, built on specific research problems, small-scale datasets, etc. Thus, when it comes to deep learning, these limitations could lead to several issues such as overfitting, and unbalanced problem.

Recently, in computer vision and natural language areas, pretrained models have generalized well to various tasks. Research showed that pretrained models can effectively reduce the training duration and improve the performance for complicated tasks such as semantic segmentation, and they can also be beneficial for tasks with small-scale datasets. Unluckily, pretrained models in audio areas are scarce, and most of them were pretrained on coarse-grained classification tasks. Because of special characteristics of audio data, nearly all pretrained methods are based on CNNs, of which waveforms and log-mel spectrograms of short-time audio clips are two common inputs.

However, pretrained audio models are still useful, because they can be used to extract embedding features, which are usually hard to be quantified with traditional methods but quite effective for specific tasks. Meanwhile, with transfer learning, pretrained models' parameters can also be fine-tuned on the new tasks.

2.4.1 Audio-based Models

Most of the pretrained audio-based models are trained on AudioSet [18]. AudioSet is a dataset collected from Youtube videos, which contains over 5,000 hours of audio recordings with more than 500 audio event classes such as human and animal sounds, musical instruments, and common everyday environmental sounds. This large-scale dataset has been one of the most potential audio sources for transfer learning systems.

VGGish [19], released by Google in 2017, is a system pretrained on initial AudioSet. As its name suggests, this model is based on VGGs, but it just has four groups of convolution/maxpool layers instead of five. VGGish uses 2-D mel spectrograms as inputs, and provides a high-level 128-D embedding layer for features extraction. VGGish can be a great "warm start" for the lower layers of audio models, and it also shows great potential to benefit audio classification tasks.

PANNs [20] is also a system pretrained on AudioSet. Compared with VGGish, PANNs also uses CNNs as the backbone, but PANNs proposes a novel architecture called Wavegram-Logmel-CNN, which takes both waveforms and mel spectrograms as inputs. In addition, PANNs can also be effectively transferred to a range of other audio classification tasks.

2.4.2 Music-based Models

Different from AudioSet, which is a kind of coarse-grained audio dataset, the MagnaTagATune Dataset [21] and the Million Song Dataset [22] are two large datasets consisted of only music tracks with multiple labels. The MagnaTagATune is a dataset collected by a two-player online music game, and it has 19k training songs with more than 188

tags such as instrumentals, emotions, music genres, etc. The Million Song Dataset is a collection of audio features and metadata for a million contemporary popular music tracks, and its annotation contains not only basic music information such as key, tempo, and loudness, but also sophisticated genre tags.

Musicnn [23] is a CNNs based library pretrained on the MagnaTagATune Dataset and the Million Song Dataset. Similar with VGGish, musicnn takes mel spectrogram as inputs, and also uses VGGs as backbone. However, musicnn take advantages of two special types of convolution kernels with shapes of $7 \times N$ and $N \times 1$ as temporal and timbrel layers. When it comes to feature extraction, even though pretrained on smaller music datasets, musicnn shows similar performance as VGGish on the GTZAN dataset. In addition, rather than the 128-D features as VGGish, musicnn provides more optional feature layers for feature extraction, so its lower layers have the potentials to be transferred to other tasks such as sentiment analysis, rather than simple music classifications.

2.5 Deep Learning Models for Sequential Data

The most famous deep neural networks that model the sequential dependency of data are RNN [24], LSTM [25], GRU [26] and Transformer [27] in the order of release time. These models assume the output of each state is dependent on the other states in the sequence. RNN takes a serialized form, where a state is processed after the states before it have been processed, besides, RNN cannot process very long sequences because of gradient vanishing and exploding problems. LSTM introduces three gates–an input gate, an output gate, and a forget gate to control the flow of information—the extent to which a new input flows into the memory, the extent to which the value remains in memory, and the extent to which the memory is used to compute the output. GRU has fewer parameters than LSTM, and reduces gates to an update gate and a reset gate. Due to their recurrent nature, RNN, LSTM, and GRU cannot be trained in parallel, Transformer however, uses self-attention mechanism to allow parallelization. Besides, Transformer can learn dependencies of two states without considering their distance in the sequence, thereby reducing the drop in performance in long sequences.

3. PROPOSED FEATURES FOR MER

In this section, we propose the lyric and audio features which we consider beneficial for music emotion recognition task, and we use these proposed features as input for our model.

3.1 Lyrics Features

Bag-of-Words and TF-IDF Bag-of-Words is a direct means of converting sentences into vectors. The only information of this method is word frequency, so we cannot infer which word appears first when using this method. TF-IDF is an intuitive method to convert sentences into vectors. Among them, TF represents the word frequency of a word in the document, and IDF represents the importance of the word in the document. Since words are sparse in lyrics, we only calculate Bag-of-Words and TF-IDF for the 5000 most frequent words.

NMF(Non-negative matrix factorization) The 5000-dimensional vector space is still relatively high in dimension and the matrix density comparison coefficient, we use NMF to reduce the 5000-dimensional Bag-of-Words vector to 300 dimensions. The advantage of using NMF is: the value of each element in the text data vector is a non-negative number, as negative number has no practical meaning, and the NMF decomposition results in a non-negative matrix, so it naturally meets this characteristic and requirement.

Emotion Feature The emotional tendency score of each word is obtained using the Afinn emotional dictionary (`https://github.com/fnielsen/afinn`). We calculate the emotion feature as sum of affin score of each word divided by the number of words.

Proportion of Part of Speech Words of different parts of speech may contain different emotions, so we take the proportion of different types of words into consideraton.

Word Vector from BERT BERT and other Transformer encoder architectures have achieved great success in various tasks of NLP (Natural Language Processing). BERT series model uses the Transformer encoder architecture to process each tag of the input text. We insert a [CLS] symbol in front of the lyric, and use the output vector of the symbol given by BERT as the semantic representation of the entire text. Compared with other words in the text, this symbol without obvious semantic information will more "fairly" integrate the semantic information of each word in the text.

3.2 Audio Features

3.2.1 Physical features

Music has the ability to evoke the emotional responses of human beings. According to research in Philosophy and Musicology areas, even though listening to music is a simple way to alter mood, there are a myriad of features that all contribute to listener's emotion [28].

Tempo The tempo is the speed or pace of a musical piece. Fast tempo can bring experiences like anger, while slow tempo can build sadness or serenity.

Loudness The loudness or the energy is the physical strength and amplitude of a sound. The powerful sound will bring feeling like anger or excitement, but a weak performance could lead to nostalgia.

Key The key or the mode is the type of scale. Major tonality usually makes people feel happy and relaxed, but minor tonality is often applied to create a sorrowful mood.

Melody The melody is the linear succession of musical tones that the listener perceives as a single entity. When the melody complement harmonies, it will bring happiness, relaxation. However, with clashing harmonies, the melody itself can build a terrifying or mournful atmosphere.

3.2.2 Audio Features Selection and Extraction

The tempo and the key of a piece of music can be easily measured by DAWs or toolkits. We use them as static features in our model.

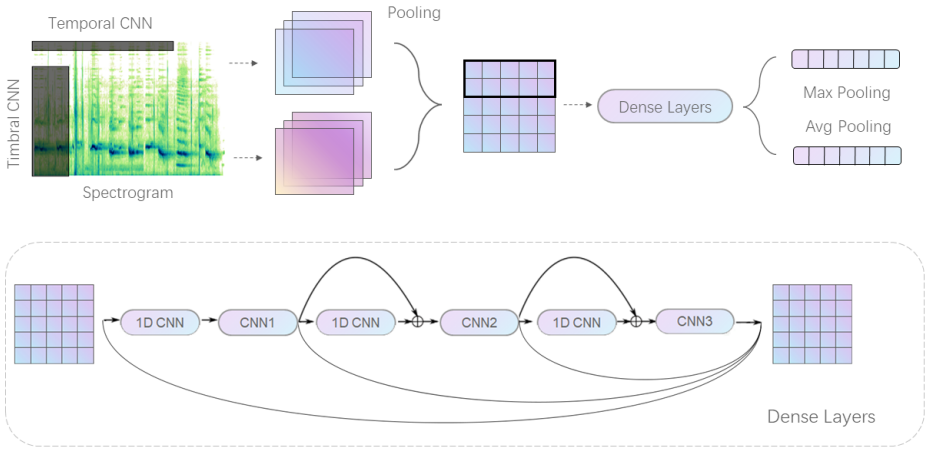

Figure 3. Framework of Musicnn

However, the melody and the energy of different levels of frequency are difficult to be quantified with only waveforms data. Spectrogram [29] is useful, but this kind of complex 2-D image-like data only works well on large datasets, or it would lead to overfitting issues. Since most of the annotated datasets in MER are limited in the number of songs, thus, pretrained models based on spectrogram are necessary for the extraction of such features. In addition, musical sentiment analysis is a fine-grained task rather than music genre classification, so models as musicnn, which pretrained on pure music datasets rather than general audio datasets, are supposed to be more useful. Inputting 1-second audio clips into musicnn, we finally extracted the 753-D mean pool layers as one of our model's inputs.

OpenSMILE [30], released in 2010, is a source-available software for features extraction from audio signals. The openSMILE toolkit includes music classification technology for automatic music mood analysis and recognition of chorus segments, key, tempo, and genre. Even with the development of deep learning, the OpenSMILE still shows a great ability to extract useful audio features as the inputs of music analysis models. Eventually, 260 dynamic features from openSMILE, which include energy related, vocal and spectral aspects, are also applied in our model, and we refer to these 260 features as music260 in the rest of this paper.

4. PROPOSED MODEL

In this section, we propose the architecture of our model which can predict the Valence-Arousal of the entire song and dynamically predict the Valence-Arousal of the song in every 1 second. As each word of lyric is hard to match each second of the music track, we treat the lyric of music as static descriptor, and the audio of music as dynamic descriptor. In terms of modeling the static feature and dynamic feature, if we concatenate the non-sequential static data to the dynamic data and feed them as a whole to a sequential model, we will pollute the sequential dynamic data, and make it harder for the model to study the inner dependencies. Therefore, we come up with the idea of dividing the model into two parts—the static part and the dynamic part. We input the static features such as BERT embeddings of lyrics and the attributes of music (i.e., key and tempo) and model them by specially designed MLP, and we input the sequential audio features such as musicnn and music260 embeddings and model them by Transformer. The final valence and arousal scores are calculated as the weighted sum of the predictions from the static part and the dynamic part using two gating networks.

We can also look at our model from another perspective, considering that each song has a relatively fixed emotion basis, and the emotion at each second fluctuates around this basis, so the static part can be regarded as learning the overall emotional basis of music, and the dynamic part can be regarded as learning the temporal emotional changes at each second. Therefore, it's intuitive to sum the results from static part and dynamic part to get the final result. Below, we discuss the components of our model in detail. The overview of our model is illustrated in Figure 4.

Modal head Modal heads are linear projections applied on embeddings from different sources, in order to reduce dimension and help different embeddings from different sources better adapt to our task. As embeddings from difference sources vary a lot in length, and some are quite long, which may demand a large number of parameters, inspired by the multi-head concept in Transformer [27], we introduce modal head, each modal head is a linear projection, which projects the embedding from corresponding source to the same length. For an embedding e_i with length d_i from source i, if we want all the embeddings to be fixed length of d_{model}, then the modal head h_i will be of shape $d_i \times d_{model}$. We apply modal heads on embeddings generated from BERT, NMF, musicnn, and openSMILE, and project them to fixed length of 50.

Static part The input of static part is the unchanged static features for a song, and we use pretrained BERT

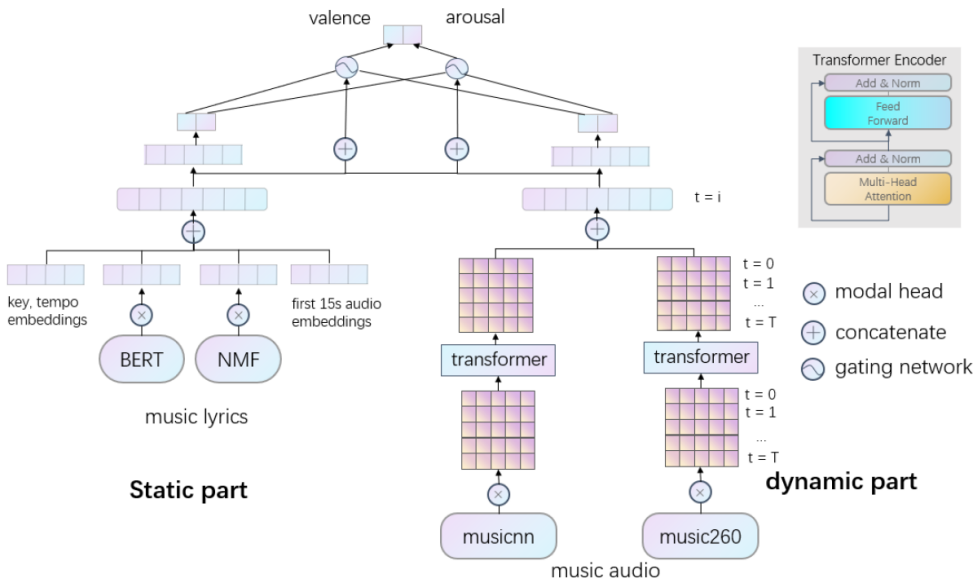

Figure 4. Overview of our proposed model for music emotion recognition

embeddings, the non-negative matrix factorization (NMF) results of Bag-of-Words model, and static attributes—the key and tempo which are transformed to learnable embeddings. Besides, there is a concept called initial orientation time (IOT) in MER that around 8-12 seconds are needed before the annotators can make meaningful and reliable annotations [31] [32]. Therefore, we introduce this concept to our model, and aggregate the first 15 seconds' embeddings from musicnn and music260 by average pooling. Next, embeddings from different sources are mapped to the same length by modal heads and concatenated and input to a two-layer MLP with batch normalization and dropout techniques. The static model finally outputs a 2-dimensional vector, which represents the valence and arousal values. Since the valence and arousal values range between 0 and 1 in the dataset, we use sigmoid as the activation function. Let $\boldsymbol{X} \in \mathbb{R}^m$ be the concatenated features, $\boldsymbol{W_1} \in \mathbb{R}^{n \times m}$ be the weight matrix, and $\boldsymbol{b_1} \in \mathbb{R}^n$ be the bias. The hidden state $\boldsymbol{h}$ is given by:

$$\boldsymbol{h} = f(\boldsymbol{W_1 X} + \boldsymbol{b_1})$$

Where f is the activation function, here we use ReLU. Let $\boldsymbol{W_2} \in \mathbb{R}^{2 \times m}$ be the weight matrix, and $\boldsymbol{b_2} \in \mathbb{R}^2$ be the bias in the second layer. And the output of the static part $output_S$ is:

$$output_S = \sigma(\boldsymbol{W_2 h} + \boldsymbol{b_2})$$

where σ is the sigmoid function, $\sigma(x) = \frac{1}{1+e^{-x}}$, and σ is applied on each dimension of its input.

Dynamic part The input of dynamic part is the embeddings for every second of the audio obtained by musicnn and openSMILE(music260) and are projected to fixed length by modal heads. Since the audio embeddings are time series data, and the emotion at current moment is strongly related to the emotion at the surrounding moments. Therefore, we use Transformer which has good performance on long-sequence data, and is highly parallelizable and fast, to learn the internal relationship of the audio sequences. As embeddings generated from musicnn and openSMILE are different time series, in practice, we find it better to use two separate Transformers to model these two sequences instead of concatenating them and use only one Transformer. In terms of the recognition of global emotion of the entire song, for each song, we concatenate an identical learnable embedding in front of the dynamic audio embeddings, and use the output at the first dimension as the representation of the entire audio. The embeddings obtained by Transformer are input into an MLP, and the output is 2-dimensional, which represents the valence and arousal values predicted by the dynamic part. Let $\boldsymbol{E_{nnt}} \in \mathbb{R}^m$ be the output of musicnn Transformer and $\boldsymbol{E_{260t}} \in \mathbb{R}^m$ be the output of music260 Transformer at time step t, $\boldsymbol{W} \in \mathbb{R}^{2 \times 2m}$ be the weight matrix, and $\boldsymbol{b} \in \mathbb{R}^2$ be the bias. Then the output of dynamic part at time step t is:

$$output_{Dt} = \sigma(\boldsymbol{W} C(\boldsymbol{E_{nnt}}, \boldsymbol{E_{260t}}) + \boldsymbol{b})$$

Where C is the concatenation operator, σ is the sigmoid function, $\sigma(x) = \frac{1}{1+e^{-x}}$, and σ is applied on each dimension of its input.

Gating network We use gating networks to calculate the weight to combine the output of static part and dynamic part. Since the input of the static model is mainly text features, the input of the dynamic model is audio features, and studies have shown that text and audio perform differently in predicting valence and arousal [5]. For example, arousal represents how calming or exciting the song is, and thus is better reflected by the audio signals than by the lyrics.

Therefore, we combine the output of the static part and the dynamic part with attention mechanism. Specifically, we build two gating networks for valence and arousal respectively. The input of gating network is the concatenation of static and dynamic features, after batch normalization and dropout, they are feed into an MLP which outputs a 2-dimensional vector, representing the weight of the static part and the dynamic part, we use softmax as the activation function. Let $\boldsymbol{X_S} \in \mathbb{R}^{m_1}$ be the static features and $\boldsymbol{X_{Dt}} \in \mathbb{R}^{m_2}$ be the dynamic features at time step t. Let $\boldsymbol{W_V} \in \mathbb{R}^{2\times(m_1+m_2)}$ be the weight matrix and $\boldsymbol{b_V} \in \mathbb{R}^2$ be the bias for the valence gate. Let $\boldsymbol{W_A} \in \mathbb{R}^{2\times(m_1+m_2)}$ be the weight matrix and $\boldsymbol{b_A} \in \mathbb{R}^2$ be the bias for the arousal gate. Then the output of the valence gate and arousal gate at time step t is:

$$Gate_{Vt} = \phi(\boldsymbol{W_V} C(\boldsymbol{X_S}, \boldsymbol{X_{Dt}}) + \boldsymbol{b_V})$$

$$Gate_{At} = \phi(\boldsymbol{W_A} C(\boldsymbol{X_S}, \boldsymbol{X_{Dt}}) + \boldsymbol{b_A})$$

Where ϕ is the softmax function, and C is the concatenation operator. Last, the predictions of valence and arousal at time step t are calculated by:

$$Valence_t = Gate_{Vt}[0]*output_S[0]+Gate_{Vt}[1]*output_{Dt}[0]$$

$$Arousal_t = Gate_{At}[0]*output_S[1]+Gate_{At}[1]*output_{Dt}[1]$$

Where $[i]$ represents the ith value of the vector. If $t = 0$, then $Valence_t$ and $Arousal_t$ are the global predictions of valence and arousal, otherwise they are the dynamic predictions of valence and arousal at time step t.

5. EXPERIMENTS

In this section, we apply our model on the PMEmo dataset, to demonstrate the effectiveness of our proposed model. Besides, experiments on the sequential model selection and feature importance are conducted.

5.1 Dataset

Music emotion recognition (MER) research requires a large music content library. Zhang [8] designed a music emotion experiment which recruits 457 experimental subjects from various countries and professions to eliminate influence of cultural and educational background. The experiment collected electrodermal activity (EDA) signals of subjects for 794 songs, with emotional annotations completed by at least 10 people for each song, and formed a high-quality multi-modal music corpus——PMEmo. The dataset is open source at www.happiness.zju.edu.cn for benchmark testing in music emotion retrieval and recognition. PMEmo also provides a pre-calculated acoustic feature set for the evaluation of music sentiment analysis methods, and provides the chorus part of the song (compressed as MP3) manually extracted by music majors students for related research.

The PMEmo data set includes the following data:

• Song metadata (song title, artist, start and end timestamp of chorus part);

• Manually excerpted music and human voice clips (MP3);

• Pre-computed acoustic feature set, and the music260 features we used are from this dataset.

• Manual emotion annotations: static annotations for the entire clip (i.e. global annotations), and dynamic annotations for each 0.5 second segment (i.e. continuous annotations over a period of time), we average annotations for every two 0.5 seconds and get the annotation for each second.

• Corresponding EDA physiological signals.

PMEmo contains a list of songs from 2016-2017 Billboard Hot 100, iTunes Top 100 (USA) and UK Top 40 Singles after deduplication, including 794 popular songs. In order to reduce the heavy cognitive burden and improve the stability of emotion annotation, MER-related data sets including PMEmo use music fragments as emotion-inducing materials. Unlike most other data sets, PMEmo does not intercept equal-length fragments, but precise and unequal-length human voice parts, which is manually extracted by music major students.

5.2 Data Process and Training

Data Process Since the annotators need initial orientation time (IOT) before they can make reliable annotations, the IOT in PMEmo dataset is 15 seconds, and songs which are shorter than 15 seconds don't have annotations, therefore, we delete the songs which are shorter than 15 seconds, then 767 songs are left. We treat the first 700 songs as the training set, and the remaining 67 songs as the test set. The longest song in the dataset lasts 88 seconds, plus the first dimension for the embedding of the whole song, therefore, we define the sequence length as 89. The first embedding of each song is the same learnable embedding, whereas embeddings in other position are fixed embeddings extracted from musicnn or openSMILE, any position which are invalid in the sequence will be assigned zero embeddings.

Training We use Mean Square Error (MSE) as loss function, at training time, we calculate global loss as MSE of global emotion annotation and the $t = 0$ dimension of the output, and also calculate the MSE of dynamic emotion of valid positions, and sum the global and dynamic MSE up as the training loss. Specifically, we use position mask, the mask of any position which is invalid in the original song is 0, and will not count towards the loss. We use Adam [33] as the optimizer with initial learning rate set as 0.001, the number of Transformer encoder layer is 2, and we use the sinusoidal function as positional encoding.

5.3 Base Models

We select commonly used machine learning models to recognize music emotions, including LASSO, Ridge regression, SVR, Random Forest and XGBoost. We use these models to fit the data and adjust the parameters to improve the performance of each model.

LASSO LASSO adds L1 regularization to the basic standard linear regression, and selects more important features from a large number of features for model prediction. Among them, the parameter alpha is the coefficient of the

regular term, and the initial value is 1. The larger the value, the greater the penalty for the complex model. Finally, an alpha value of 0.002 is selected and applied to the model.

Ridge regression Ridge regression utilizes L2 regularization. An alpha value of 25 is selected and applied to the model.

SVR SVR is a variant of SVM. It aims at obtaining a surface or a function to maximize its "distance" to the farthest sample point of the hyperplane, so as to fit the data. An important parameter of SVR is the kernel function, which essentially implies a low-dimensional to high-dimensional mapping, thereby avoiding direct calculation of the high-dimensional inner product. We select the Gaussian kernel function, and the value of C is 0.5.

Random Forest Random forest ensemble multiple trees to train and predict samples. $Min_samples_leaf$ is the minimum number of samples of leaf nodes. Smaller leaves make it easier for the model to capture the noise in the training data. $N_estimators$ is the number of sub-models. More sub-trees can make the model have better performance, but at the same time make the code running iteration slow. Finally, $min_samples_leaf$ is set as 100 and $n_estimators$ is set as 100.

XGBoost XGBoost improves the traditional GBDT, effectively prevents over-fitting and improves computational efficiency. The parameter values are as follows:

$$colsample_bytree = 0.8, gamma = 0.045,$$

$$learning_rate = 0.07, max_depth = 8,$$

$$min_child_weight = 1.5, n_estimators = 300,$$

$$reg_alpha = 0.65, re_lambda = 0.45, subsample = 0.95.$$

5.4 Comparison with Other Models

Evaluation Metrics We use Root Mean Square Error (RMSE) and Pearson Correlation Coefficient (r) as evaluation metrics as they were used in the PMEmo paper [8]. These two metrics are calculated separately for global and dynamic emotion recognition. In addition, for dynamic emotion recognition, we use the average of RMSE and r for each song as the RMSE and r for dynamic recognition.

We compare our model with the base models mentioned in the PMEmo paper [8], and also with our own base models. For global emotion, the performance of PMEmo models are from its paper. For dynamic emotion, the performance of PMEmo models are from its github, and there are no records of r on dynamic emotion recognition. The performance on global emotion recognition is in Table 1, and the performance on dynamic emotion recognition is in Table 2.

The results show that our model has the lowest RMSE among all the models in both global emotion recognition and dynamic emotion recognition. The global emotion recognition performance in Table 1 shows that our model decreases the RMSE of PMEmo MLR by 25.0% in valence, and 16.2% in arousal. The highest r in arousal is achieved by our model while the highest r in valence is achieved by our base SVR model. According to Table 2, Our model preforms the best on dynamic emotion recognition, as the metrics on dynamic recognition are higher than those of global recognition. Overall, we observe an improvement in RMSE of 43.4% for valence and 43.7% for arousal compared with the base PMEmo MLR. Besides, most of our base models achieve better performance than the base PMEmo models, which shows that our proposed features are useful for the music emotion recognition task.

Our model performs especially well on the dynamic emotion recognition, this may due to the fact that for the global emotion recognition, the input global embedding is the same for each song, after output by Transformer, the global embedding changes according to the local embeddings of the song. Nevertheless, it makes the global recognition a little bit harder than the dynamic recognition, as the input of dynamic embeddings are the authenticate values, while the input of global embeddings are the same.

5.5 Model Selection

We change the Transformer in the dynamic part to RNN, LSTM and GRU to compare the performance of these sequential models. Also, we experiment with a base model (denoted as None) that does not model the sequential relationship between the audio features, by directly inputting the original audio embeddings to the MLP in the dynamic part. Since RNN, LSTM and GRU are recurrent models, therefore, we cannot evaluate their performance on the global emotion recognition as the first embedding of different songs are the same. The results of different models on dynamic emotion recognition are shown in Table 3.

The results show that Transformer is the best among all the models, and LSTM and RNN are close to each other, and the performance of GRU is worse than that of RNN and LSTM. Also, it is necessary to model the sequential relationship of audio clips, as all of the sequential models perform better than the base without sequential model. The good performance of Transformer may result from its better ability for long sequence and its bidirectional modelling mechanism. We considered about using masked multi-head attention for Transformer, where the input only attends to the former input. However, as emotion of one episode is strongly related to the emotion of surrounding episodes, plus the annotators will listen to each music clip over and over again to give annotations for each second, therefore, we think it's more reasonable to use the bidirectional attention.

5.6 Feature Importance

We conduct experiments to compare the feature importance of BERT, NMF, musicnn and music260 embeddings for music emotion recognition by ablation experiment. We remove one of these embeddings each time, and compare the performance. The results in Table 4 show that audio features like musicnn and music260 are more important than the lyric features, since without musicnn or music260, the performance drops significantly.

Besides, we can examine the importance of lyric and audio by comparing the gating network output for valence and arousal. The average of gating networks, namely, the average weight of static part and dynamic part are give

Table 1. Performance on global emotion recognition

Model	RMSE(Valence)	RMSE(Arousal)	r(Valence)	r(Arousal)
ours	**0.102**	**0.093**	0.738	**0.837**
PMEmo MLR	0.136	0.111	0.546	0.719
PMEmo SVR	0.124	0.102	0.638	0.764
Lasso	0.125	0.103	0.566	0.815
Ridge	0.134	0.134	0.569	0.731
SVR	0.123	0.101	**0.757**	0.691
RF	0.110	0.100	0.646	0.602
XGBoost	0.110	0.099	0.660	0.620

Table 2. Performance on dynamic emotion recognition

Model	RMSE(Valence)	RMSE(Arousal)	r(Valence)	r(Arousal)
ours	**0.090**	**0.076**	**0.813**	**0.895**
PMEmo MLR	0.159	0.135	-	-
PMEmo SVR	0.120	0.117	-	-
Lasso	0.118	0.118	0.696	0.776
Ridge	0.123	0.123	0.744	0.802
SVR	0.119	0.117	0.810	0.842
RF	0.119	0.117	0.810	0.810
XGBoost	0.118	0.116	0.808	0.883

Table 3. Sequential model comparison on dynamic emotion recognition

Model	RMSE(Valence)	RMSE(Arousal)	r(Valence)	r(Arousal)
Transformer	**0.090112**	**0.075949**	**0.702722**	**0.814628**
RNN	0.098503	0.081847	0.619854	0.794329
LSTM	0.099796	0.081475	0.681596	0.805889
GRU	0.121553	0.111018	0.573151	0.704438
None	0.140700	0.122320	0.528972	0.660352

Table 4. Ablation experiment on feature importance

Model	RMSE(Valence)	RMSE(Arousal)	r(Valence)	r(Arousal)
origin	0.090112	0.075949	0.702722	0.814628
Without BERT	0.109592	0.082274	0.638598	0.763183
Without NMF	0.101404	0.094627	0.677216	0.747631
Without musicnn	0.109323	**0.114297**	**0.500554**	**0.466623**
Without music260	**0.110402**	0.097355	0.641304	0.761295

Table 5. Model comparison on dynamic emotion prediction

Model	RMSE(Average)	local r(V.)	local r(A.)	global r(V.)	global r(A.)
MiT	0.125	**0.098**	**0.108**	0.550	0.660
PMEmo MLR	**0.106**	0.071	0.104	**0.674**	**0.830**
PMEmo EDA	0.165	0.060	0.098	0.040	0.026

Table 6. Average weight of static part and dynamic part

	static part	dynamic part
Valence	0.383	0.617
Arousal	0.369	0.631

in Table 6. The results show that the weight of the static part is smaller than that of the dynamic part, further proving that the audio features are more important than the lyric features in our task. Besides, comparing valence and arousal, the dynamic part weigh more than the static part for arousal than for valence, which is consistent with R. Delbouys et al. [5], as arousal is closely related to energy, which are essentially reflected by the audio signal, while valence can be explained by both lyric and audio.

Because of the importance of audio features, we also attempt a pretrained audio model (MiT) based on ViT [34], whose input is only images. Inspired by ViT, we split each spectrogram into patches and provide the sequence of linear embeddings of both spectrograms and patches as an input of Transformer Encoder. Simply, patches are treated the same way as words in an NLP application, and each spectrogram is treated as a sentence. After the Encoder, two linear layers and the softmax function are added for training on classification tasks.

When it comes to pretraining, we train our MiT model on the FMA-medium [35] dataset (20,000 music clips) and save its encoder's weights. Specially, we set a very large length as 1024 for the position embeddings of spectrograms and patches. Even trained on a medium dataset, our MiT model shows a pleasant result (Top1 accuracy: *60%*) on the GTZAN dataset.

Then we transfer this pretrained model to the dynamic emotion recognition task by modifying its linear layers and also adding a sigmoid function before the output. Using Adam as the optimizer and the dynamic MSE as loss, we fine-tune our pretrained MiT model on the PMEmo dataset. With only spectrograms as input, our model still shows an acceptable performance (Table 5), especially for arousal recognition. If we could pretrain it on a larger dataset, it is entirely possible to combine this MiT model with the static part of our main model.

6. CONCLUSION

We propose a multi-modal deep learning model to recognize the global and dynamic emotion of music, based on lyrics and audio signals. The model is composed of two parts, the static part which inputs static features such as key, tempo, and BERT embeddings of lyrics, and the dynamic part which inputs sequential embeddings extracted from pretrained musicnn model and openSMILE. We use the first sequential output as the global emotion and the other sequential outputs as dynamic emotion for each second. Attention mechanism is adopted to combine the predictions of the static part and the dynamic part by introducing gating networks. The experiment results show that our model is the best among all the other models and it is especially good at recognizing the dynamic motion. Besides, we find that audio features are more important for the accuracy of music emotion recognition.

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